

MS6721: SUPPLY CHAIN MANAGEMENT

Effective Term

Semester B 2025/26

Part I Course Overview

Course Title

Supply Chain Management

Subject Code

MS - Decision Analytics and Operations

Course Number

6721

Academic Unit

Decision Analytics and Operations (DAOS)

College/School

College of Business (CB)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

MS6721A Supply Chain Management

Exclusive Courses

FB6721 Supply Chain Management

Part II Course Details

Abstract

This course provides a comprehensive overview of supply chain management, using the textbook "Supply Chain Management: Strategy, Planning, and Operation" as a foundation. Students will explore key concepts and strategies essential

for effective supply chain management in various industries. The course begins with an introduction to supply chain management principles and progresses through critical topics such as supply chain drivers, inventory management, and risk pooling. Students will learn about push-pull strategies, the intricacies of supply contracts, and the complexities of transportation and distribution network design. Additionally, the course covers demand forecasting techniques and the bullwhip effect, emphasizing their impact on supply chain efficiency. By the end of the course, students will have a robust understanding of how to design, manage, and optimize supply chains to enhance operational performance and meet strategic business objectives.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if DEC-A1 app.)		DEC-A2	DEC-A3
1	Identify the scope of supply chain management and the major sources of challenges in supply chain management.	20	x		
2	Align appropriate supply chain strategies with product characteristics to enhance efficiency and effectiveness.	30		x	x
3	Select and justify performance measures that meet both organizational and customer needs.	10		x	x
4	Utilize Information Technology and objective alignment theory to address and resolve supply chain coordination issues.	20		x	x
5	Develop operational strategies to effectively mitigate risks within the supply chain.	20		x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Interactive Lectures and Discussions	Participate in interactive lectures and discussions to identify the scope and challenges of supply chain management, and to align supply chain strategies with product characteristics, stimulating curiosity, critical thinking, and active questioning	1, 2	3

2	Case Study Analyses and Group Projects	Engage in case study analyses and contribute to group projects where we select and justify performance measures, and develop risk mitigation strategies for hypothetical supply chain scenarios, fostering critical thinking, real-life application, and creativity	2, 3, 5	1
3	Lab Sessions	Utilize supply chain management software and information technology tools in lab sessions to address and resolve coordination issues, and to develop operational strategies, applying theory to practice and demonstrating effective solutions	4, 5	1

Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks (e.g. Parameter for GenAI use)
1	Case Exercise	1, 2, 3	50	
2	Assignments	4, 5	20	

Continuous Assessment (%)

70

Examination (%)

30

Examination Duration (Hours)

2

Assessment Rubrics (AR)**Assessment Task**

Case Exercise (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The ability to analyze real-world supply chain cases, identify the scope and challenges of supply chain management, align strategies with product characteristics, and justify performance measures.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The ability to apply course concepts and theories to develop solutions for supply chain issues.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The ability to identify supply chain scope and challenges, align strategies with product characteristics, select and justify performance measures, utilize IT and theory for coordination, and develop risk mitigation strategies.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Case Exercise (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The ability to analyze real-world supply chain cases, identify the scope and challenges of supply chain management, align strategies with product characteristics, and justify performance measures.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Basic

Failure

(F) Not even reaching marginal levels

Assessment Task

Assignments (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The ability to apply course concepts and theories to develop solutions for supply chain issues.

Excellent

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Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The ability to identify supply chain scope and challenges, align strategies with product characteristics, select and justify performance measures, utilize IT and theory for coordination, and develop risk mitigation strategies.

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Part III Other Information

Keyword Syllabus**Introduction to Supply Chain Management (SCM)**

What is SCM? The complexity and the key issues in SCM. Supply chain strategies.

Logistics Network

Key parties and their responsibilities. Data essential for logistics network configuration.

Inventory Management

Demand forecasting. Strategic sourcing. Distribution requirement planning, material requirement planning. Inventory management, risk pooling and postponement, and distribution strategies for the supply chain.

Information and Supply Chain Management

The value of information. Bullwhip Effect. Beer Game. Information for supply chain integration and coordination. Vendor managed inventory.

Global supply chain management

Issues such as tax/tariff and rules of origin/trade agreements pertinent to global supply chains; risk management; globalization strategies

Special topics in SCM

Supply chain contract; revenue management; smart pricing.

Technologies and SCM

Enabling technologies such as IT and Internet of Things/RFID and SCM. Emerging trends in the industry.

Reading List**Compulsory Readings**

Title	
1	David Simchi-Levi, Philip Kaminsky, Edith Simchi-Levi, Designing and Managing the Supply Chain, McGraw-Hill, 2009 (3rd ed.)
2	Gerald Cachon and Christian Terwiesch, Matching Supply with Demand, McGraw-Hill, 2013 (3rd ed.)

Additional Readings

Title	
1	Sunil Chopra, Peter Meindl, Supply Chain Management, Strategy, Planning & Operations, Pearson, 2007.